Pharmacology Exam Sheet

Therapeutic Uses & Adverse Effects · 6 lectures · quick revision

How to study: read each table → cover the adverse-effects column and quiz yourself → review the high-yield box → finish with the "Which drug causes…?" page. **Bold** = most examinable. Content is taken only from your 6 lecture slides.

Lecture 1 · The Haemopoietic System (Anticoagulants, Antiplatelets, Fibrinolytics)

Lecture 2 · Drugs Affecting Coagulation, Bleeding & Thrombosis (Coagulants)

Lecture 3 · Drugs for Anaemia

Lecture 4 · Antimalarial Drugs

Lecture 5 · Chemotherapy for Leishmaniasis

Lecture 6 · Anticancer Drugs

Lecture 1 · The Haemopoietic System (Anticoagulants, Antiplatelets, Fibrinolytics)

Anticoagulants

Drug	Therapeutic uses ✓	Adverse effects ⚠
Heparin & LMWHs (IV / SC — not absorbed orally)	• Treatment of acute **DVT** and **pulmonary embolism** • Prophylaxis of **post-operative venous thrombosis** (elective surgery) • Prevent clotting in **extracorporeal devices** (e.g. dialysis machines)	• **Haemorrhage** • Hypersensitivity reactions • **Thrombocytopenia**
Warfarin (oral, Vit K antagonist)	• Prevent progression/recurrence of **DVT or PE** after initial heparin • Prevent VTE during **orthopaedic / gynaecologic surgery** • Prophylaxis in **acute MI**, **prosthetic heart valves**, **chronic atrial fibrillation**	• **Bleeding disorders** • Crosses placenta → haemorrhagic disorder in fetus • **Teratogenic** → abortion & birth defects → **NEVER in pregnancy**

Antiplatelet drugs

Drug	Therapeutic uses ✓	Adverse effects ⚠
Aspirin	• Prophylaxis of **transient cerebral ischaemia (TIA)** • Reduce **recurrent MI** • Decrease mortality in pre- & post-MI patients	*Not mentioned in the lecture*
Dipyridamole	• Prophylaxis of **angina pectoris** • Given **with aspirin or warfarin** (ineffective alone)	*Not mentioned in the lecture*
Ticlopidine	• Prevent **TIA & stroke** in patients with prior cerebral thrombotic event • With aspirin after **coronary stent** → ↓ stent thrombosis	• **Neutropenia / agranulocytosis** • **TTP** (thrombotic thrombocytopenic purpura) • **Aplastic anaemia**
Clopidogrel	• Prevent atherosclerotic events after recent **MI, stroke**, or established **peripheral arterial disease**	*Not mentioned in the lecture*

Fibrinolytics & haemostatics

Drug	Therapeutic uses ✓	Adverse effects ⚠
Streptokinase, Alteplase, Duteplase (plasminogen → plasmin)	• **DVT** and serious **pulmonary embolism** • Dissolve clots causing **stroke**	• **Haemorrhage** • **Contraindicated:** healing wounds, pregnancy, history of CVA (stroke)
Tranexamic acid · Protamine · Vit K · Aprotinin	• **Tranexamic acid**: inhibits plasminogen activation → prevents fibrinolysis • **Protamine sulfate**: **antidote of heparin** (+ charge binds − heparin) • **Aprotinin**: serine-protease inhibitor → blocks plasmin → stops bleeding • **Vitamin K**: haemostatic (see Lecture 2)	*Not mentioned in the lecture*

★ High-yield exam points

- Heparin = fast, injectable (IV/SC). Warfarin = oral, Vit K antagonist (factors II, VII, IX, X).
- Antidotes: **Heparin → Protamine** · **Warfarin → Vitamin K**.
- Pregnancy: **warfarin NEVER** (teratogenic); thrombolytics contraindicated.
- Ticlopidine's AEs are all **blood** problems: neutropenia, TTP, aplastic anaemia.

Lecture 2 · Drugs Affecting Coagulation, Bleeding & Thrombosis (Coagulants)

Vitamin K

Drug	Therapeutic uses ✓	Adverse effects ⚠
Vitamin K K1 Phytonadione · K3 Menadione	• Dietary deficiency • Prolonged **antimicrobial** therapy (kills gut flora) • **Obstructive jaundice** / malabsorption • **Antidote for warfarin bleeding** • **Liver disease** (cirrhosis, viral hepatitis) • **Newborns** • Prolonged high-dose **salicylate** → hypoprothrombinaemia	• **Hypersensitivity / anaphylactoid reactions**

Miscellaneous coagulants

Drug	Therapeutic uses ✓	Adverse effects ⚠
Fibrinogen (human)	• Bleeding in **haemophilia** • **AHG deficiency** • Acute **afibrinogenaemic** states	*Not mentioned in the lecture*
Antihaemophilic factor (concentrated human AHG)	• **Haemophilia** • **AHG deficiency**	*Not mentioned in the lecture*
Desmopressin (releases factor VIII + vWF)	• Bleeding in **haemophilia** • **von Willebrand's disease**	*Not mentioned in the lecture*
Adrenochrome monosemicarbazone	• ↓ capillary fragility, oozing from raw surfaces • **Epistaxis**, **haematuria**, secondary haemorrhage from wounds	*Not mentioned in the lecture*
Ethamsylate (platelets must be adequate; NOT antifibrinolytic)	• Capillary bleeding in **menorrhagia**, after abortion • Epistaxis, **malena**, haematuria, after **tooth extraction**	*Not mentioned in the lecture*

Antifibrinolytics

Drug	Therapeutic uses ✓	Adverse effects ⚠
EACA & Tranexamic acid (block lysine site on plasminogen; TXA 10× more potent → preferred)	• Bleeding due to **fibrinolytic drugs** • **Cardio-pulmonary bypass** surgery • **Tonsillectomy, prostatic surgery, tooth extraction** in haemophiliacs • **Menorrhagia**, esp. due to **IUCD** • Recurrent **epistaxis**, **hyphema** (ocular trauma), **peptic ulcer**	• Nausea & diarrhoea • **Thromboembolic events** • **Disturbed colour vision**, allergic reactions • **Thrombophlebitis** of injected vein

★ High-yield exam points

- Vit K is a cofactor for liver synthesis of **II, VII, IX, X** → useful only when bleeding is due to deficiency of these factors.
- Desmopressin works by releasing **factor VIII + vWF** from endothelium.
- Ethamsylate needs **adequate platelets** and does NOT stabilise fibrin.
- Antifibrinolytics stop bleeding → so their big risk is **thrombosis**.

Lecture 3 · Drugs for Anaemia

Iron

Drug	Therapeutic uses ✓	Adverse effects ⚠
Oral iron Ferrous sulfate / fumarate / gluconate, polysaccharide–iron, carbonyl iron	• **Iron-deficiency anaemia** (hypochromic microcytic) • Dose in lecture: **150–180 mg/day** elemental iron in 2–3 divided doses	• **GI disturbances** (local irritation): abdominal pain, constipation, diarrhoea • **Dark stools** = MOST common AE
Parenteral iron Iron dextran, sodium ferric gluconate, iron sucrose	• Iron-deficiency anaemia	• **Fatal hypersensitivity / anaphylactoid reactions** (mainly **iron dextran**)

Vitamins & growth factors

Drug	Therapeutic uses ✓	Adverse effects ⚠
Folic acid	• **Megaloblastic anaemia** • **Prophylaxis in pregnancy** (1 mg/day) → ↓ **neural tube defects** • **Methotrexate toxicity** • Enhance anticancer efficacy of **5-FU**	*Not mentioned in the lecture*
Vitamin B12 Cyanocobalamin, hydroxocobalamin	• **Treatment of B12 deficiency** (e.g. pernicious anaemia – no intrinsic factor) • Prophylaxis **only** when definite predisposing factors exist • Mega-doses in neuropathies, psychiatric disorders, cutaneous sarcoid, as a tonic → **value questionable**	*Not mentioned in the lecture*
Erythropoietin & Darbepoetin (darbepoetin = long-acting)	• Anaemia of **end-stage renal disease** • Anaemia with **HIV** infection • Anaemia in **bone marrow disorders** • **Anaemia of prematurity** • Anaemia in some **cancer** patients	*Not mentioned in the lecture*

★ High-yield exam points

- Match drug to anaemia type: **Iron → microcytic** · **Folate / B12 → megaloblastic (macrocytic)** · **EPO → renal anaemia**.
- Oral iron: most common AE = **dark stools**. Parenteral iron: most dangerous = **anaphylaxis (iron dextran)**.
- EPO is made by the **kidney** → that's why its #1 use is anaemia of **renal failure**.
- Careful: high-dose methotrexate rescue is **leucovorin (folinic acid)**, not plain folic acid (Lecture 6).

Lecture 4 · Antimalarial Drugs

Antimalarials

Drug	Therapeutic uses ✓	Adverse effects ⚠
Primaquine	• Eradicates **primary exo-erythrocytic (liver/tissue)** forms • Eradicates secondary exo-erythrocytic forms of **P. vivax & P. ovale → ONLY drug that prevents relapses** • Not active on erythrocytic stage → combine with **chloroquine / mefloquine**	• **Haemolytic anaemia in G6PD deficiency** • **Not in pregnancy** • Resistance can develop in all Plasmodium species
Chloroquine	• **Drug of choice** for erythrocytic **P. falciparum** (except resistant strains) • Less effective against P. vivax • **Prophylaxis** for travel to chloroquine-sensitive areas	• GI upset • **Pruritus** (itching) • Headache & **blurred vision**
Atovaquone–proguanil	• **Chloroquine-resistant P. falciparum** • **Prevention & treatment** of malaria • Take with **food or milk** (↑ absorption)	• Nausea, vomiting, abdominal pain • Headache, diarrhoea, anorexia, dizziness
Mefloquine (t½ ≈ 20 days)	• **Single agent** for prophylaxis & treatment of **multidrug-resistant P. falciparum**	• High doses: nausea, vomiting, dizziness • → **disorientation, hallucinations, depression** (neuropsychiatric)
Quinine (cinchona bark)	• **Severe** malaria • **Chloroquine-resistant** strains • Given with **doxycycline, tetracycline or clindamycin**	• AEs not listed. Interaction: absorption ↓ by **aluminium antacids**
Artemisinin (sweet wormwood)	• **First-line** for **multidrug-resistant P. falciparum** • **Never used alone** (to prevent resistance)	• Nausea, vomiting, diarrhoea
Pyrimethamine	• Blood schizonticide + strong sporonticide • Not alone for P. falciparum → fixed combination with **sulfadoxine**	*Not mentioned in the lecture*

★ High-yield exam points

- **Relapse (vivax/ovale) → Primaquine** (only one). Its key AE → **haemolysis in G6PD deficiency**.
- Sensitive falciparum → **chloroquine**. Resistant → **atovaquone-proguanil, quinine**. Multidrug-resistant → **mefloquine, artemisinin**.
- Neuropsychiatric AE (hallucinations, depression) → **Mefloquine**. Blurred vision + pruritus → **Chloroquine**.
- "Never alone": **artemisinin**, **pyrimethamine** (+ sulfadoxine), **primaquine** (+ blood schizonticide).

Lecture 5 · Chemotherapy for Leishmaniasis

Antileishmanial drugs

Drug	Therapeutic uses ✓	Adverse effects ⚠
Sodium stibogluconate (SSG) (deep IM into buttocks / slow IV)	• **First-line** for **visceral leishmaniasis (VL)** — > 90% cure • Accumulates in macrophages → prolonged effect	• Nausea, vomiting, **metallic taste**, cough, abdominal pain • **Pain & stiffness of injected muscle** • **Pancreatitis**, liver & kidney damage, myelosuppression (seldom severe)
Amphotericin B AMB-DOC · Liposomal L-AMB (binds ergosterol)	• **Second-line** for VL • **Mucocutaneous** leishmaniasis	*Not mentioned in the lecture*
Miltefosine (interferes with membrane phospholipids → apoptosis)	• **First ORAL drug** for visceral leishmaniasis • Some activity in cutaneous & mucocutaneous forms	• **Nausea & vomiting** (common) • **Teratogenic → avoid in pregnancy**
Paromomycin	• **Oral**: intestinal protozoal infections (stays in gut) • **IM**: VL (approved in India 2006) • Effective in **SSG-resistant** cases • WHO: **21-day** course as alternative to miltefosine	*Not mentioned in the lecture*

★ High-yield exam points

- VL: **1st line = SSG** · **2nd line = Amphotericin B** · **Oral = Miltefosine** · **SSG-resistant = Paromomycin**.
- Metallic taste + injection-site pain + pancreatitis → **SSG**.
- Teratogenic → **Miltefosine** (avoid in pregnancy).

Lecture 6 · Anticancer Drugs

Alkylating agents (cell-cycle non-specific)

Drug	Therapeutic uses ✓	Adverse effects ⚠
Mechlorethamine	• **Hodgkin's** & non-Hodgkin's lymphoma	• **Bone marrow depression** (dose-limiting) • Nausea/vomiting, alopecia, diarrhoea, **sterility**
Cyclophosphamide / Ifosfamide	• Lymphoma, **breast**, ovarian cancer, sarcoma • Some **immune conditions**	• **Haemorrhagic cystitis** (from **acrolein**) → prevent with **MESNA** + hydration
Nitrosoureas Carmustine, Lomustine (lipophilic → cross BBB)	• **Brain tumours** (primary & metastatic) • Hodgkin's & NHL • Adenocarcinoma of stomach, colon, rectum • Hepatocarcinoma	• **DELAYED bone marrow depression** • **Pulmonary fibrosis**
Busulfan	• **Chronic granulocytic (myeloid) leukaemia**	• Bone marrow depression • **Pulmonary infiltrates & fibrosis** • Nausea/vomiting, alopecia
Platinum: Cisplatin, Carboplatin, Oxaliplatin	• **Testicular**, ovarian, lung, bladder, colorectal cancers	• **Nephrotoxicity** • **Ototoxicity** • Neuropathy • **Severe nausea**

Antimetabolites (S-phase specific)

Drug	Therapeutic uses ✓	Adverse effects ⚠
Methotrexate (DHFR inhibitor)	• Leukaemia, lymphoma • **Osteosarcoma**, **choriocarcinoma** • Some **autoimmune** diseases (RA, psoriasis)	• **Mucositis** • **Marrow suppression** • **Hepatotoxicity** • Low dose → give **folic acid**; High dose → **leucovorin rescue**
6-Mercaptopurine (6-MP)	• **ALL** (main use) • Some **AML** regimens • Immunosuppression in **IBD**	• Bone marrow suppression • **Hepatotoxicity** • Nausea, vomiting, anorexia • Mucositis / stomatitis
Azathioprine (prodrug → 6-MP; immunosuppressant)	• **Organ transplant rejection** prevention • **Rheumatoid arthritis** • **IBD** (Crohn's, UC) • Autoimmune hepatitis, **SLE**, vasculitis	• Bone marrow suppression • Hepatotoxicity • Nausea & vomiting • **Pancreatitis**
5-Fluorouracil (5-FU) (thymidylate synthase inhibitor)	• **Colorectal**, gastric, pancreatic, breast, head & neck cancers • **Topical**: actinic keratosis & superficial skin cancers	• Bone marrow suppression, mucositis, **diarrhoea**, N/V • **Hand-foot syndrome**, alopecia • Rare: **cardiotoxicity** • **DPD deficiency** → severe/fatal toxicity

Antitumour antibiotics

Drug	Therapeutic uses ✓	Adverse effects ⚠
Doxorubicin	• **Breast** cancer, lymphomas, sarcomas, leukaemias	• **Dose-related cardiomyopathy** • Myelosuppression • Nausea, vomiting, diarrhoea
Bleomycin	• **Hodgkin** lymphoma, **testicular** cancer, squamous tumours	• **Pulmonary fibrosis** • **Little myelosuppression**

Drug	Therapeutic uses ✓	Adverse effects ⚠
Mitomycin C	• With vincristine for **breast cancer**	• Myelosuppression

Plant alkaloids

Drug	Therapeutic uses ✓	Adverse effects ⚠
Vincristine & Vinblastine (M-phase, block tubulin polymerisation)	• Leukaemias, lymphomas, selected solid tumours	• **Vincristine → neuropathy** • **Vinblastine → more myelosuppression**
Paclitaxel & Docetaxel (stabilise microtubules)	• Breast, ovarian, lung, prostate cancers	• Myelosuppression • **Peripheral neuropathy** • **Hypersensitivity reactions**
Etoposide (topoisomerase II inhibitor)	• **Testicular** cancer, **small-cell lung** cancer	• Myelosuppression • Vomiting, alopecia, diarrhoea

Hormonal therapy

Drug	Therapeutic uses ✓	Adverse effects ⚠
Tamoxifen (SERM – blocks breast ER)	• **Breast cancer** (hormone-dependent)	• CNS: depression, dizziness, headache, ↓ **visual acuity & retinopathy** • GI: nausea, vomiting • **Hypercalcaemia** • **Vaginal bleeding**/discharge, menstrual irregularities
Aromatase inhibitors Anastrozole, Letrozole, Exemestane	• **Breast cancer** (block estrogen synthesis)	• Hot flashes • **Arthralgia** • **Osteoporosis** → ↑ fracture risk
GnRH agonists Leuprolide, Goserelin, Triptorelin	• **Prostate** cancer • Breast cancer • Endometriosis	• Hot flashes, ↓ libido, erectile dysfunction • Vaginal dryness, amenorrhoea • Osteoporosis • **Tumour flare**
GnRH antagonists Degarelix, Relugolix	• **Advanced prostate** cancer • Endometriosis	• Hot flashes, ↓ libido, erectile dysfunction • Osteoporosis • (No tumour flare listed)

Monoclonal antibodies (-mab)

Drug	Therapeutic uses ✓	Adverse effects ⚠
Rituximab (CD20)	• B-cell lymphomas, **CLL**	• Infusion & allergic reactions (all mAbs)
Trastuzumab (HER2)	• **HER2+ breast** & gastric cancers	• **Cardiotoxicity**
Bevacizumab (VEGF)	• Colorectal, lung & other cancers	• **Hypertension**, **bleeding**, **impaired wound healing**
Cetuximab (EGFR)	• Colorectal, head & neck cancers	*Not mentioned in the lecture*
Pembrolizumab / Nivolumab (PD-1)	• **Melanoma**, lung cancer; renal (nivolumab)	• Pembrolizumab: **immune-mediated hepatitis, colitis, pneumonitis**
Daratumumab (CD38)	• **Multiple myeloma**	*Not mentioned in the lecture*

★ High-yield exam points

- **Pulmonary fibrosis** → Bleomycin, Busulfan, Nitrosoureas.
- **Cardiotoxicity** → Doxorubicin, Trastuzumab, (5-FU rare).
- **Neuropathy** → Vincristine, Taxanes, Platinum. **Nephro- + ototoxicity** → Cisplatin.
- **Haemorrhagic cystitis** → Cyclophosphamide/Ifosfamide → antidote **MESNA**.
- Least myelosuppression: **Bleomycin** & **Vincristine**. Delayed: **Nitrosoureas**.
- Brain tumours → **Nitrosoureas** (cross BBB). Choriocarcinoma → **Methotrexate**. ALL → **6-MP**. CML → **Busulfan**.

Final Review: Which drug causes…?

Cross-links between lectures — classic exam questions

Effect / topic	Drugs
Contraindicated / avoid in pregnancy	Warfarin, Primaquine, Miltefosine, Thrombolytics
Bleeding / haemorrhage	Heparin, Warfarin, Thrombolytics, Bevacizumab
Thrombosis (clots)	Antifibrinolytics (Tranexamic acid, EACA)
Thrombocytopenia / TTP	Heparin (thrombocytopenia) · Ticlopidine (TTP)
Haemolysis in G6PD deficiency	Primaquine
Hypersensitivity / anaphylaxis	Iron dextran (fatal), Vitamin K, Heparin, Taxanes, mAbs, Antifibrinolytics
Pulmonary fibrosis	Bleomycin, Busulfan, Nitrosoureas
Cardiotoxicity	Doxorubicin, Trastuzumab, 5-FU (rare)
Neuropathy	Vincristine, Paclitaxel/Docetaxel, Platinum
Hepatotoxicity	Methotrexate, 6-MP, Azathioprine, SSG
Pancreatitis	Azathioprine, SSG
Eye / vision problems	Chloroquine (blurred), Antifibrinolytics (colour vision), Tamoxifen (retinopathy)
Psychiatric (hallucinations, depression)	Mefloquine, Tamoxifen (depression)
Osteoporosis	Aromatase inhibitors, GnRH agonists & antagonists
Kidney + ear toxicity	Cisplatin
Antidotes	Heparin → Protamine · Warfarin → Vit K · High-dose MTX → Leucovorin · Cyclophosphamide → MESNA · Fibrinolytic bleeding → Tranexamic acid / EACA